

TECHNICAL MEMORANDUM

To: Dana Truscott, Director of Corporate Real Estate, Ardsley Financial Group
From: Dean Farago, PE — Principal
Date: March 9, 2011
Re: Corbin Tower Levels 12–19 — independent assessment of slab moisture condition
File: PBS-2011-0338

1. Scope of engagement

Ardsley Financial Group engaged Prewitt Building Science to provide an independent opinion on whether the epoxy moisture vapor mitigation system specified in Section 09 05 61 is necessary for the scheduled floor coverings on Levels 12 through 19.

2. Testing performed

Anhydrous calcium chloride testing in accordance with **ASTM F1869**, “Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride.” Nine test locations, one per floor on Levels 12 through 19 plus one duplicate on Level 15. Kits placed March 1, 2011 and recovered March 4, 2011 after 72 hours. Ambient conditions held at 70 to 75°F and 40 to 50% RH for 48 hours prior to placement, as the method requires.

3. Results

Kit	Level	MVER (lb / 1,000 SF / 24 hr)
CC-1	12	2.9
CC-2	13	3.1
CC-3	14	3.4
CC-4	15	3.2
CC-4D	15 (duplicate)	3.3
CC-5	16	3.4
CC-6	17	3.0
CC-7	18	2.9
CC-8	19	3.2

Range: 2.9 to 3.4 lb / 1,000 SF / 24 hr.

4. Discussion

The carpet tile pressure sensitive adhesive scheduled for this Project publishes a maximum moisture vapor emission rate of **5.0 lb / 1,000 SF / 24 hr**, and carpet tile is the majority of the

floor area. Every location tested falls below that limit with margin. We note for completeness that the resilient tile and heat-welded resilient sheet products publish a lower limit of 3.0 lb, and that five of our nine results are at or above 3.0.

We are aware that the design team's testing agency has reported in-situ relative humidity of 92 to 96 percent by ASTM F2170. We do not dispute those readings. We would observe that the two methods measure different quantities and that it is the calcium chloride figure that appears in the flooring manufacturers' own written limits.

5. Opinion

On the basis of the calcium chloride results, it is our opinion that the slabs on Levels 12 through 19 are within tolerance for the scheduled adhered floor coverings and that the epoxy moisture vapor mitigation system specified in Section 09 05 61 is **not required** for this installation.

6. Limitations

This memorandum is prepared for the exclusive use of Ardsley Financial Group. It addresses the moisture condition of the substrate as measured on the dates stated and does not address the long-term performance of the floor covering assembly. Prewitt Building Science is not the Architect of Record and does not assume responsibility for the Contract Documents.

Dean Farago
Dean Farago, PE
Principal, Prewitt Building Science